

Major Project Presentation on

App-Based Audiometer for Hearing Loss Detection and Prediction Using Machine Learning

in

Partial Fulfillment of Final Year Computer Engineering Course- Project Stage-I

by

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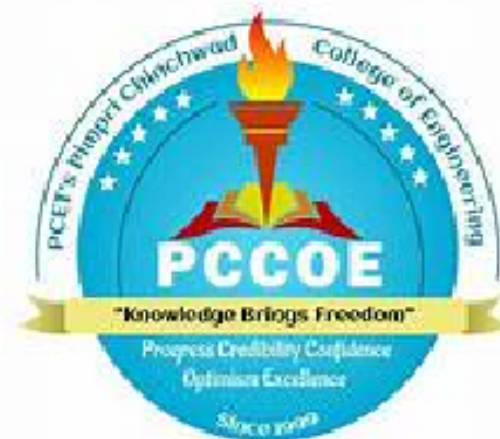
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Introduction

- **Global Hearing Loss Challenge:** WHO projects over 430 million people will suffer from disabling hearing loss by 2030, rising to 700 million by 2050. Early detection is essential to prevent negative effects on cognition, education, and quality of life.
- **Limitations of Traditional Audiometry:** Conventional pure-tone audiometry (PTA) is accurate but costly, requires trained personnel, and is often inaccessible in low-resource or rural settings.
- **Need for Mobile Solutions:** Mobile devices offer a scalable alternative for hearing assessment, but challenges include device variability, background noise, and the need for proper calibration and interface design to ensure test accuracy.
- **App-Based Audiometer (ABA) Innovation:** This research proposes an Android-based ABA, capable of both individual assessments and large-scale screenings, improving accessibility and efficiency, especially in school and community settings.
- **Integration of Machine Learning:** ML models like XGBoost, SVM, RFC, and Naïve Bayes can predict hearing loss using diverse datasets, supporting a shift from reactive to proactive care and offering societal benefits such as reduced healthcare costs and improved productivity.

Problem Statement

App-Based Audiometer for Hearing Loss Detection and Prediction Using Machine Learning

Hearing loss affects millions worldwide, yet traditional audiometry tests are costly, time-consuming, and inaccessible in many areas. There is a need for a portable, user-friendly, and cost-effective solution for hearing assessment.

Our project introduces an App-Based Audiometer that conducts hearing tests. The audiogram Data collected is loaded in Machine Learning Model.

Motivation

- **Rising Hearing Issues** – Increasing cases of hearing impairments.
- **Early Detection & Prevention** – Timely identification of hearing issues can prevent further deterioration.
- **Affordability & Scalability** – A digital, low-cost alternative to conventional audiometers, making hearing tests accessible to a larger population.
- **ML powered Analysis** – Enhancing accuracy and enabling proactive healthcare decisions through machine learning.

Scope & Feasibility

- **Scope:**

- Smartphone-Based Hearing Tests – Enables hearing assessments without the need for specialized hardware.
- Cost-Effective & Accessible – Provides an affordable solution, particularly for rural and underserved areas.
- User Data Management – Stores test results securely for future analysis and tracking.
- Early Detection & Intervention – Helps identify hearing impairments at an initial stage to prevent worsening conditions.

- **Feasibility:**

- Technical Feasibility – Uses existing smartphone hardware (Headphones, Microphones) with cloud-based processing for efficiency.
- Economic Feasibility – Reduces dependency on expensive audiometers, making hearing tests more affordable.
- Operational Feasibility – Simple app-based testing allows easy adoption without requiring specialized training.
- Market Feasibility – Growing demand for telehealth solutions ensures practical adoption and sustainability.

Objectives

- To design and develop a **Mobile Application** for conducting hearing tests using smartphone hardware, eliminating the need for traditional audiometry equipment.
- To evaluate the accuracy, sensitivity, and specificity of the developed solution, ensuring its reliability as an alternative to traditional audiometer.
- To collect, integrate, and analyze hearing test data for generating data-driven insights and tracking hearing health over time.
- To develop a Machine learning model for analyzing and comparing hearing issues in the left and right ear, identifying patterns and trends to enable better healthcare interventions.

Project Requirements

- **Software Requirements**

- Operating System – Windows/macOS (development), Android/iOS (deployment)
- Programming Language – Dart (Flutter for frontend), Python (ML model)
- Database – Firebase Realtime Database / Firestore for storing hearing test data
- Development Tools – Android Studio, VS Code, Flutter SDK

- **Hardware Requirements**

- Smartphone (Android/iOS) – For conducting and testing hearing assessments
- Microphone & Headset– Essential for generating and detecting calibration tone and BG Noise
- Computer/Laptop – Required for application development and testing
- Cloud Server (if applicable) – For data storage and processing

Project Requirements

- **Functional Requirements**

- User Authentication – Secure login and profile management
- Hearing Test Execution – Generates tones and records user responses
- Data Storage & Analysis – Stores and processes hearing test data
- Report Generation – Provides users with hearing health insights

- **Non-Functional Requirements**

- Performance – Ensures smooth and fast operation on various devices
- Usability – Simple and accessible interface for all users
- Security – Protects user data and ensures privacy
- Scalability – Supports multiple users and potential future enhancements

Literature Survey

Reference 1	Assessing the accuracy and reliability of application-based audiometry for hearing evaluation { Seung Yeol Lee, Hee Won Seo, Seon Min Jung, Seung Hwan Lee, and Jae Ho Chung , March 28, 2024 }
Objectives	To evaluate the accuracy, reliability, and feasibility of smartphone application-based audiometry compared to conventional pure-tone audiometry, assessing its effectiveness in hearing evaluation, particularly for individuals with normal hearing, while identifying its limitations in cases of hearing loss or asymmetric hearing loss.
Proposed Solution	Study proposes a smartphone application-based audiometry (Care4Ear) as a cost-effective alternative to conventional audiometry, demonstrating strong accuracy in normal hearing detection but limitations in assessing asymmetric and severe hearing loss compared to traditional pure-tone audiometry (PTA)
Results	The study achieved a 97.4% accuracy in detecting normal hearing (Class A, <35 dB), but lower accuracies of 22.2% for moderate hearing loss (Class B, 35–64 dB) and 38.5% for severe hearing loss (Class C, >65 dB).
Advantages	• Cost-effective and accessible for large-scale hearing screening. • Provides high accuracy (97.4%) for normal hearing detection. • Portable and convenient for remote or home-based evaluations. • Reduces the need for specialized equipment and trained professionals.
Limitations	• Low accuracy for moderate (22.2%) and severe (38.5%) hearing loss. • Unreliable for asymmetric hearing loss cases. • Requires a quiet environment for accurate results. • Variability in smartphone hardware and earphones affects test consistency.

Reference 2	Reliability of an Android-Based Free Application for Assessment of Hearing Loss in Adults { Pavitra Saxena and Dilip Raghavan , 2024 }
Objectives	To evaluate the reliability of the "Hearing Test" mobile application in assessing hearing levels by comparing its results with Pure-Tone Audiometry (PTA) and determining its effectiveness as a screening tool for hearing loss in a diverse patient group.
Proposed Solution	Conduct a comparative study of hearing thresholds between PTA and the Hearing Test app. Statistical analyses and diagnostic performance, were used to assess agreement and accuracy between the two methods.
Results	The app showed moderate to strong correlation with PTA at higher frequencies (2-4 kHz). However, it underestimated hearing loss by 12.5-21.98 dB, had low sensitivity (29%), but demonstrated high specificity (100%) in accurately detecting normal hearing.
Advantages	• User-Friendly: No audiologist required. • High Specificity (100%): Accurately detects normal hearing. • Fast & Convenient: Quick screening process. • Remote Accessibility: Useful in rural areas.
Limitations	• Low Sensitivity (29%): Misses some hearing loss cases. • Device & Calibration Issues: Results vary with different phones. • Ambient Noise Affects Accuracy: Needs a quiet environment. • Self-Administered Test: Users may manipulate results.

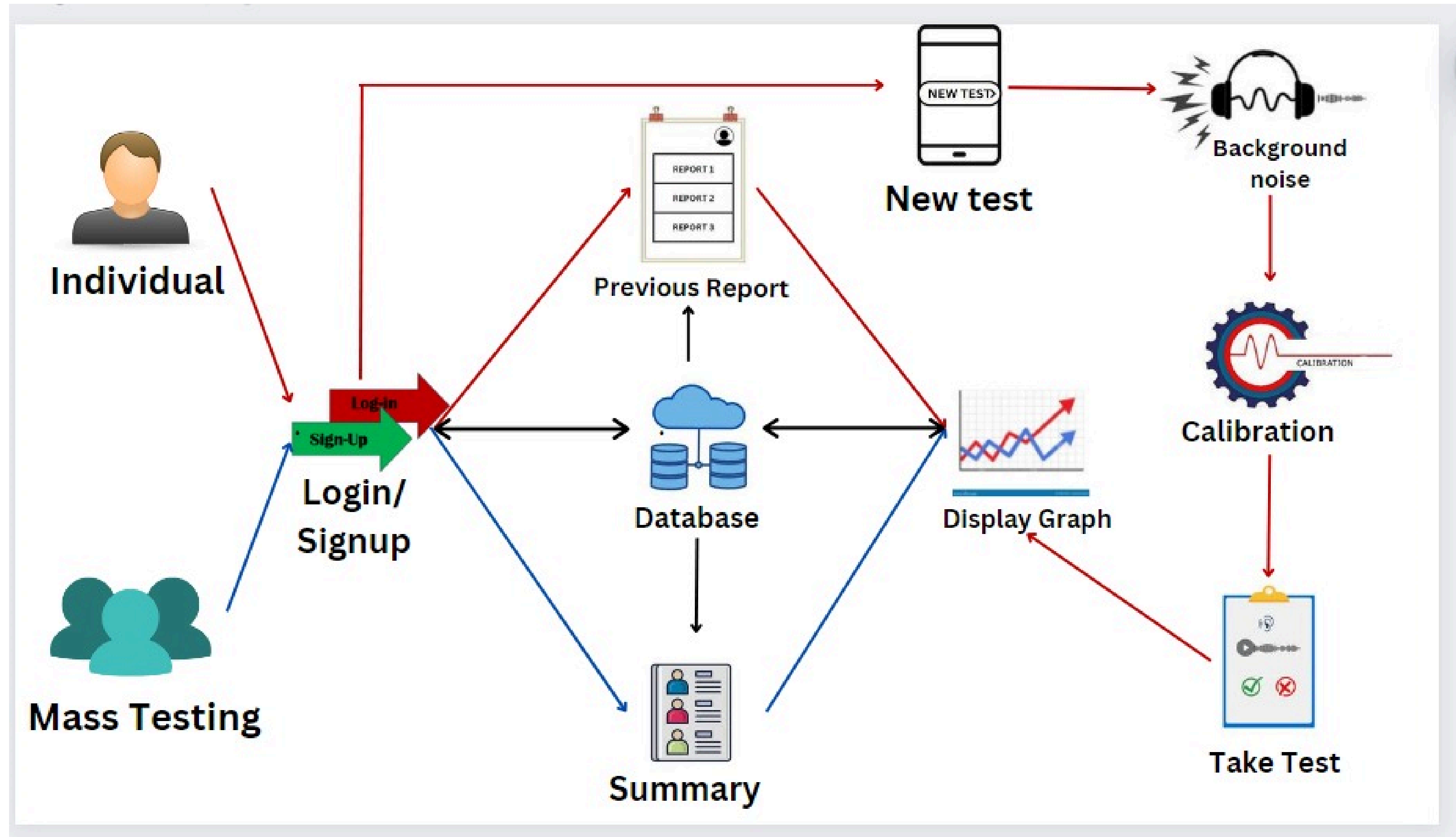
Reference 3	Hearing Loss Assessment and Analysis of Hearing Impaired Subjects Using a Facile Audiometric Technique { Dr. Sadagopan Rajkumar, S. Muttan, Vishnu Jaya, and Vignesh S. S. , 2015 }
Objectives	Develop a computer-based audiometer for hearing loss detection and audiometric data analysis. The goal is to create a cost-effective, accurate, and user-friendly system for early screening and mass diagnosis, making hearing assessments more accessible, even in remote or noisy environments.
Proposed Solution	The system is a computer-embedded expert audiometer tested on 256 subjects and validated against a standard audiometer. It offers automated hearing loss detection, result analysis, and data storage, making it useful for clinics, workplaces, and remote areas needing reliable audiometric assessments.
Results	The system showed high accuracy (90.2%), sensitivity (93.8%), and specificity (82.2%) in detecting hearing loss.
Advantages	• Early hearing loss detection before major deterioration. • High sensitivity and accuracy, ensuring reliable assessments. • Cost-effective and accessible, ideal for mass screenings. • Data storage and retrieval for long-term clinical analysis.
Limitations	• Needs larger-scale validation across diverse populations. • Relies on a computer, reducing portability. • Sensitive to environmental noise, affecting test accuracy. • Limited integration with existing clinical workflows.

Reference 4	HEAR LOSS PREDICTION USING MACHINE LEARNING { K. K. Archana, M. Bharath, U. D. Prasanna, and N. S. J. Kumar , 2024 }
Objectives	- To create a predictive tool that assesses the likelihood of hearing loss based on demographic (age) and health-related (physical score) variables. - To improve early detection and intervention for hearing loss.
Proposed Solution	A machine learning model using Gradient Boosting predicts hearing loss based on age and a "physical score(genral health, noise exposure, medical history)." The model achieved high accuracy (around 92-93%) in testing, enabling earlier detection and potential intervention.
Results	- The Gradient Boosting Classifier which attempted to predict hearing loss based on the given features, was trained using the training set that achieved a training accuracy of 93.625% and a testing accuracy of 92.6 % - The model demonstrated high levels of sensitivity, specificity, and accuracy in determining the risk of hearing loss (though the specific values aren't provided).
Advantages	- Early Detection: The system aims to identify individuals at risk of hearing loss earlier than traditional methods, enabling timely intervention. - Improved Accessibility: The user interface makes the prediction tool more accessible to a wider audience.
Limitations	- Limited Data Details - No Calibration

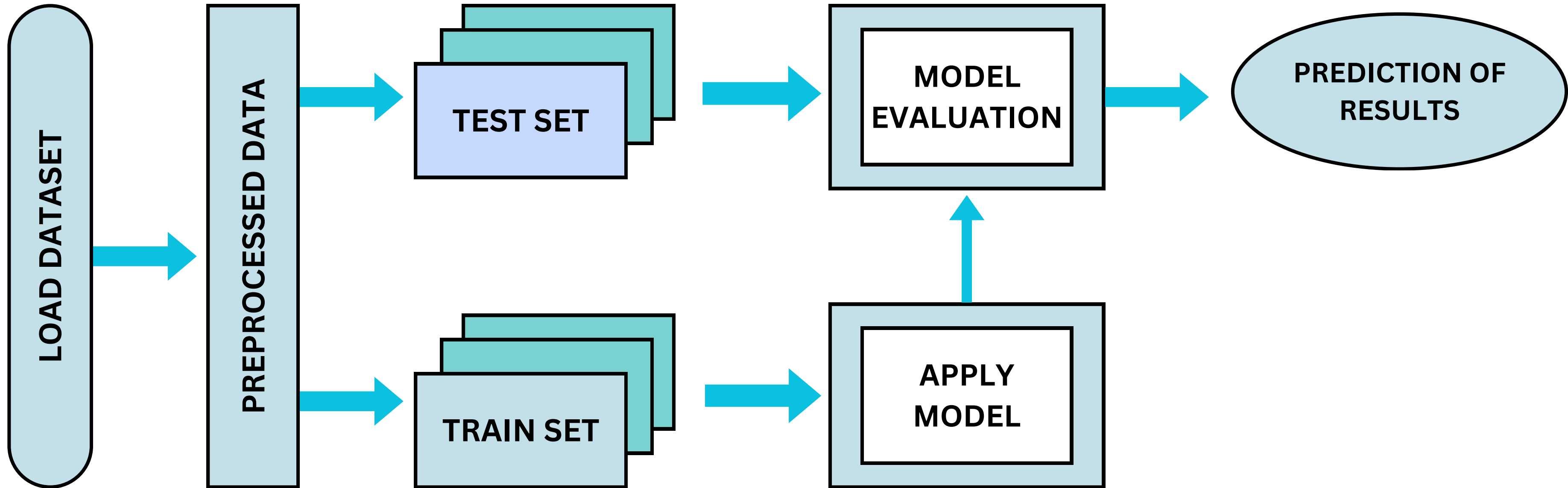
Reference 5	Machine Learning Models for the Hearing Impairment Prediction in Workers Exposed to Complex Industrial Noise { Zhao, Yanxia; Li, Jingsong ¹ ; Zhang, Meibian; Lu, Yao ¹ ; Xie, Hongwei; Tian, Yu ¹ ; Qiu, Wei ³ , June 2019 }
Objectives	To develop machine learning models for predicting noise-induced hearing loss (NIHL) in workers exposed to complex industrial noise.
Proposed Solution	1,113 workers from 17 factories underwent audiometric tests and noise exposure recording, with hearing loss classified. Machine learning models (classification and regression) analyzed key factors like age, noise exposure, and kurtosis to predict hearing impairment with 10-fold cross-validation.
Results	RF:-79.7% ; Adaboost:-78.6% ; MLP:-78.9% ; SVM:-80.1%
Advantages	• Early Identification of Hearing Loss: Helps prevent long-term damage by detecting hearing loss early. • Automated Prediction: Machine learning models provide an efficient and scalable approach to hearing health assessment.
Limitations	• Limited Dataset: The study only included workers from 17 factories, which may not generalize to all populations. • Noise Variability: Differences in workplace noise environments may affect model accuracy.

Proposed System

Architecture Diagram Of Application



ML Model Diagram



Dataset

- The dataset comprised a total of 1,000 users. Initially, data from 25 individuals was validated at an ENT clinic to ensure accuracy by comparing results from traditional audiometry tests with those from the proposed App-Based Audiometer (ABA) tests.
- Upon successful validation, the remaining data was collected from diverse locations to ensure wide demographic representation. The distribution of participants was as follows: gardens (20%), colleges (10%), residential societies (20%), hospitals (8%), public awareness campaigns (32%), and other sources (10%).
- Each data entry included parameters such as age, gender, and detailed hearing test results. The hearing tests measured the minimum intensity at which users could perceive tones at different frequencies. The frequencies tested were 250 Hz, 500 Hz, 1000 Hz, 2000 Hz, 4000 Hz, and 8000 Hz, with intensity levels ranging from 10 dB to 80 dB, in 10 dB increments.

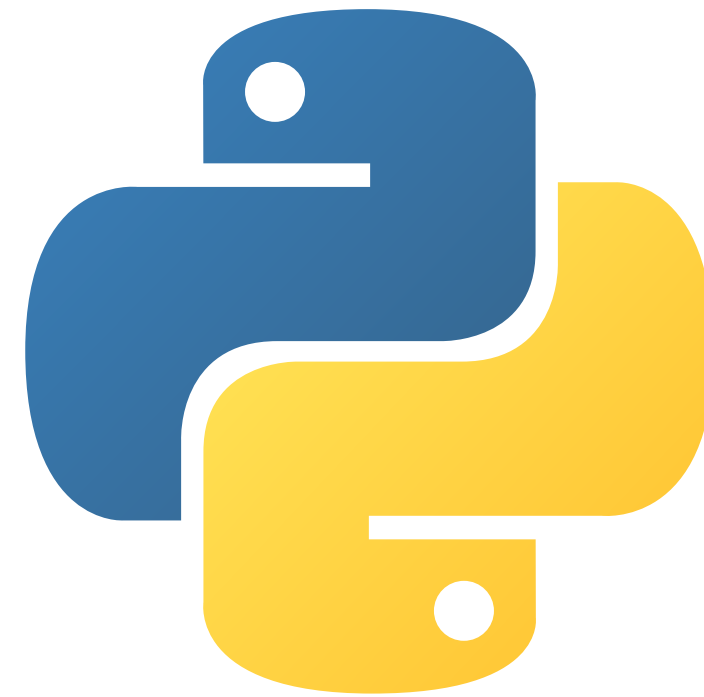
Modern Tools & Techniques selected for Project Development



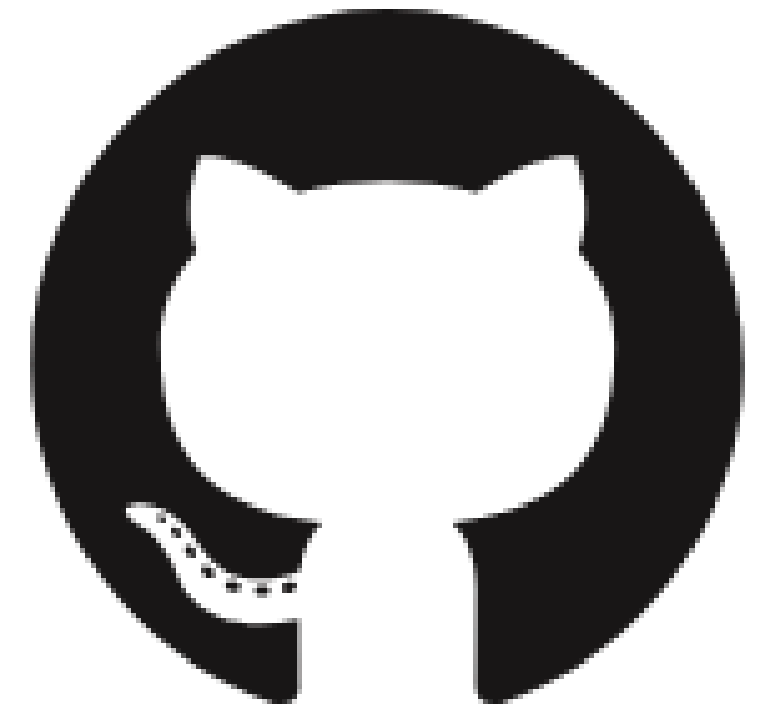
Flutter



Firebase



Python



Github

Project Plan

Week	Sprint	Task Highlights
1–2	Sprint 1	Defined project scope, problem statement, feasibility analysis, resource planning, team roles
3–4	Sprint 2	UI/UX wireframe design in Figma, user journey mapping, initial prototype using Ui-Zard
5–6	Sprint 3	Flutter project setup, navigation and UI screen development, VelocityX UI integration
7-8	Sprint 4	Firebase Realtime Database setup, Sign-In authentication, data schema and model design
9-10	Sprint 5	Audiometer logic implementation, tone generation, threshold calibration UI, Firebase integration
11-13	Sprint 6	ML model development for hearing classification, test data evaluation, sensitivity/specificity
14-15	Sprint 7	Unit testing and integration testing.
16-17	Sprint 8	Pilot testing with real users, feedback collection, bug fixes, UI refinements
18-20	Sprint 9	Final testing, documentation, user manual, and submission preparation

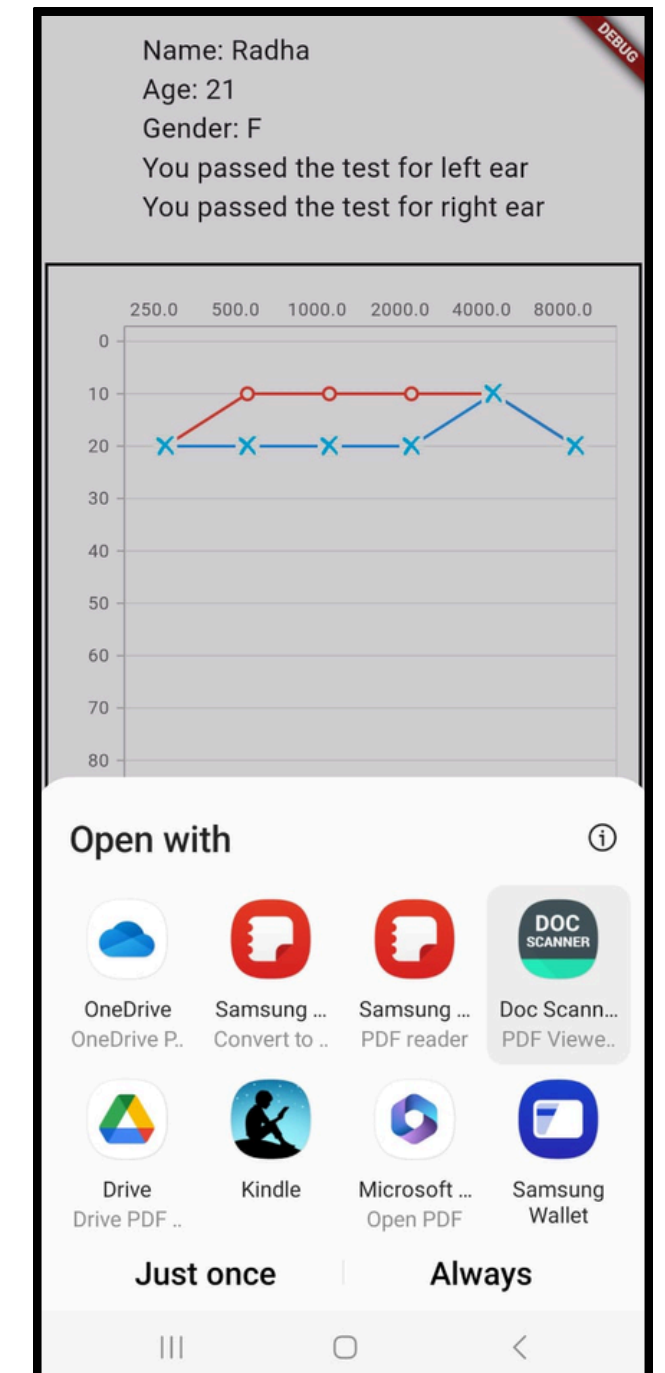
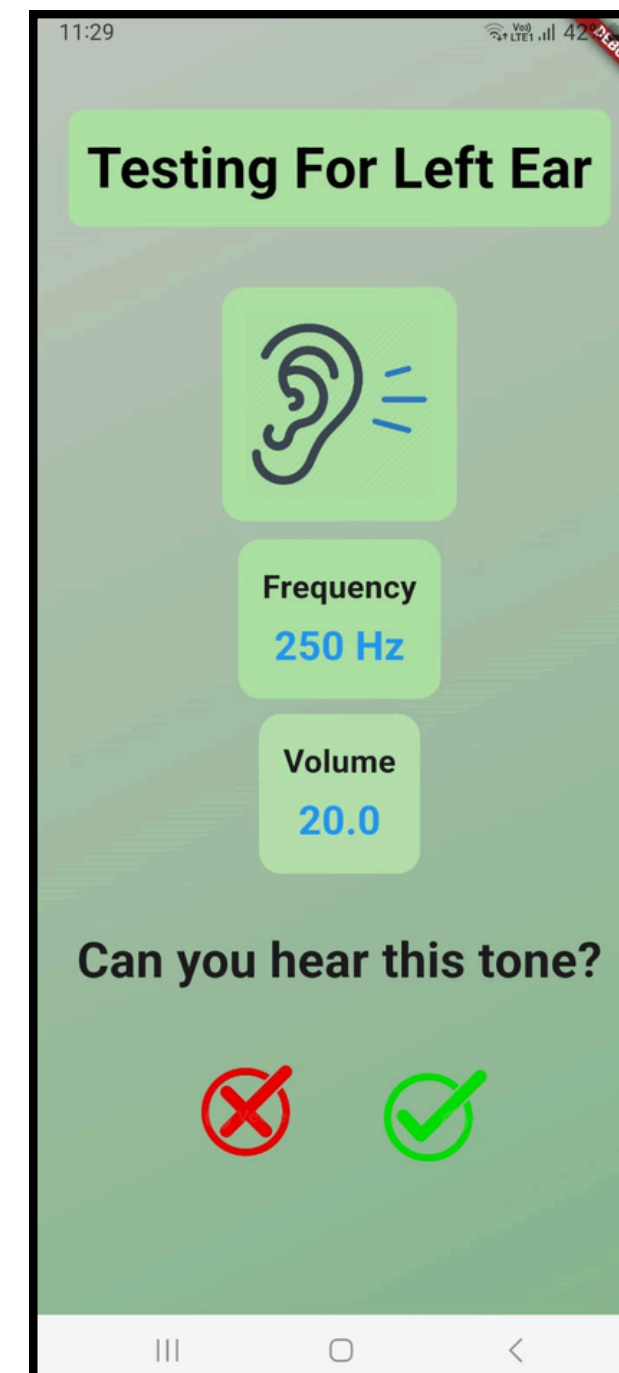
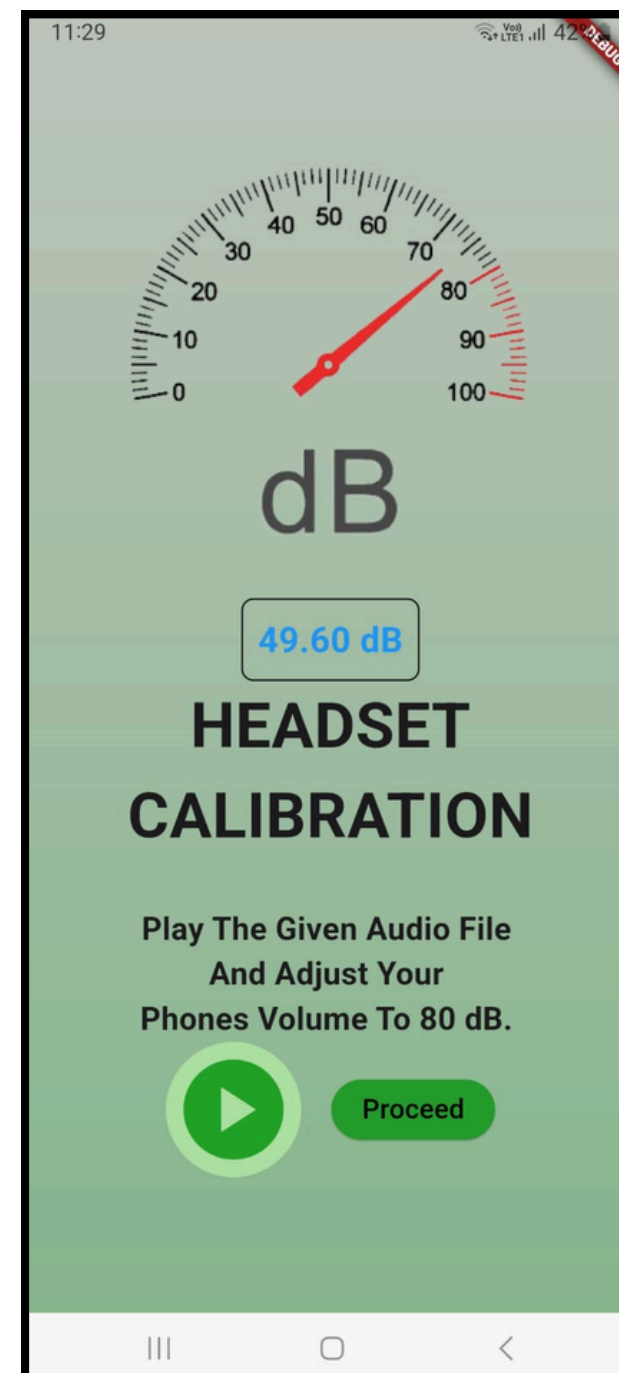
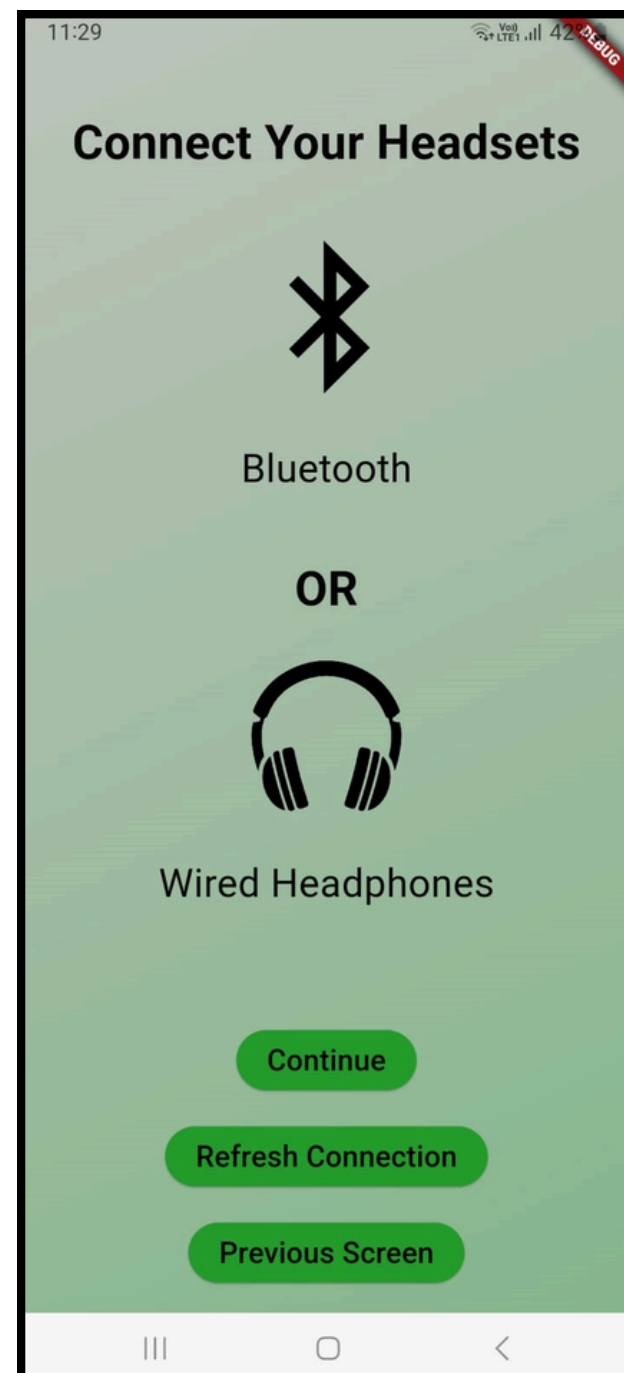
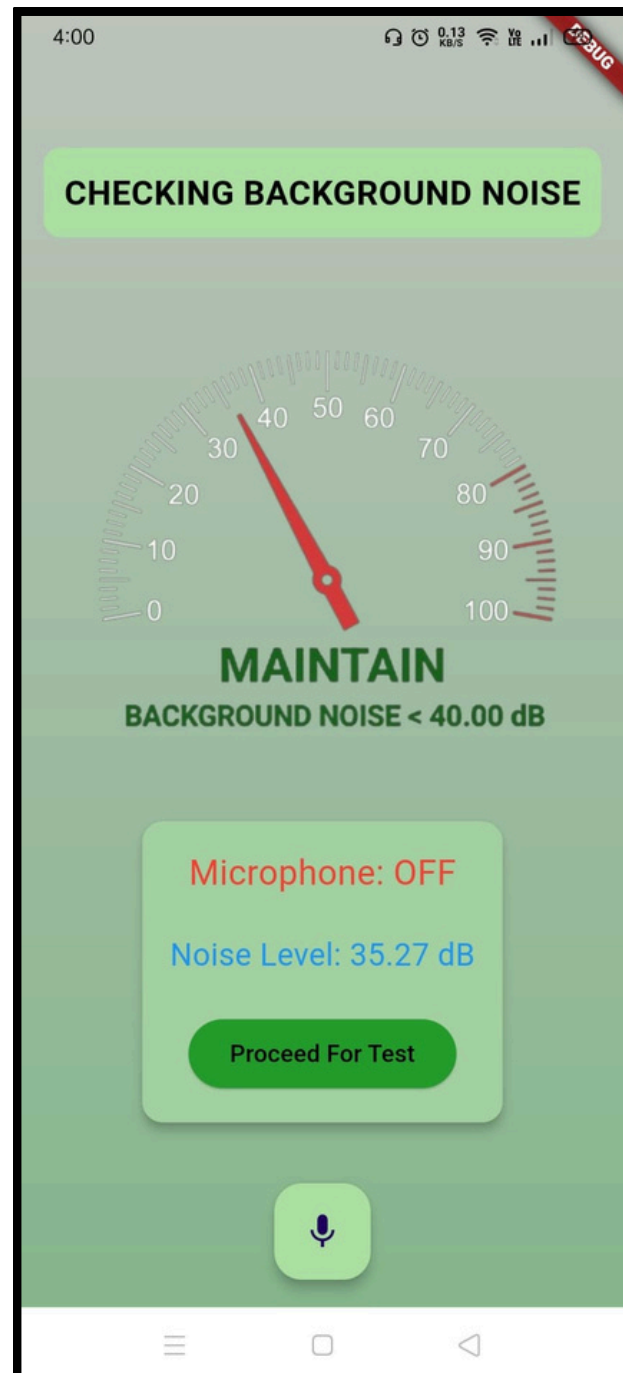
Software Testing

Testing Types:

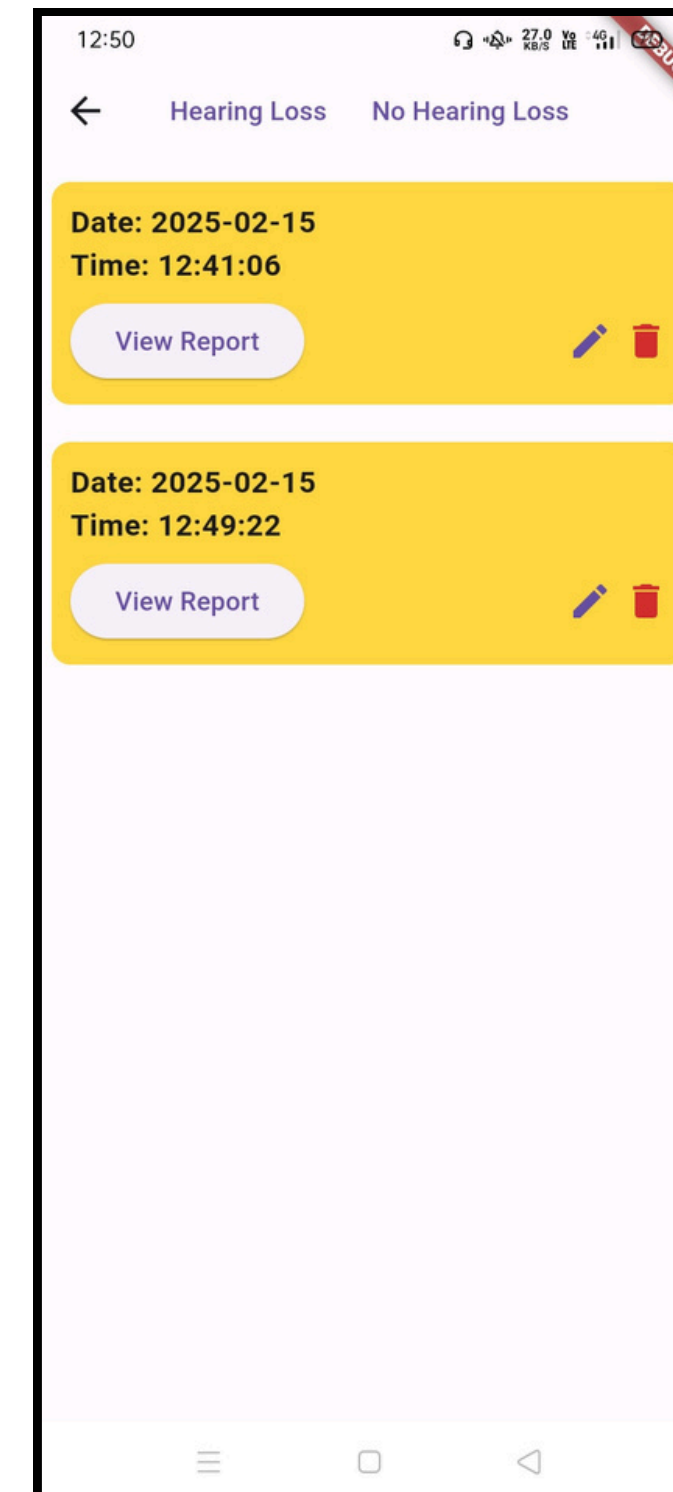
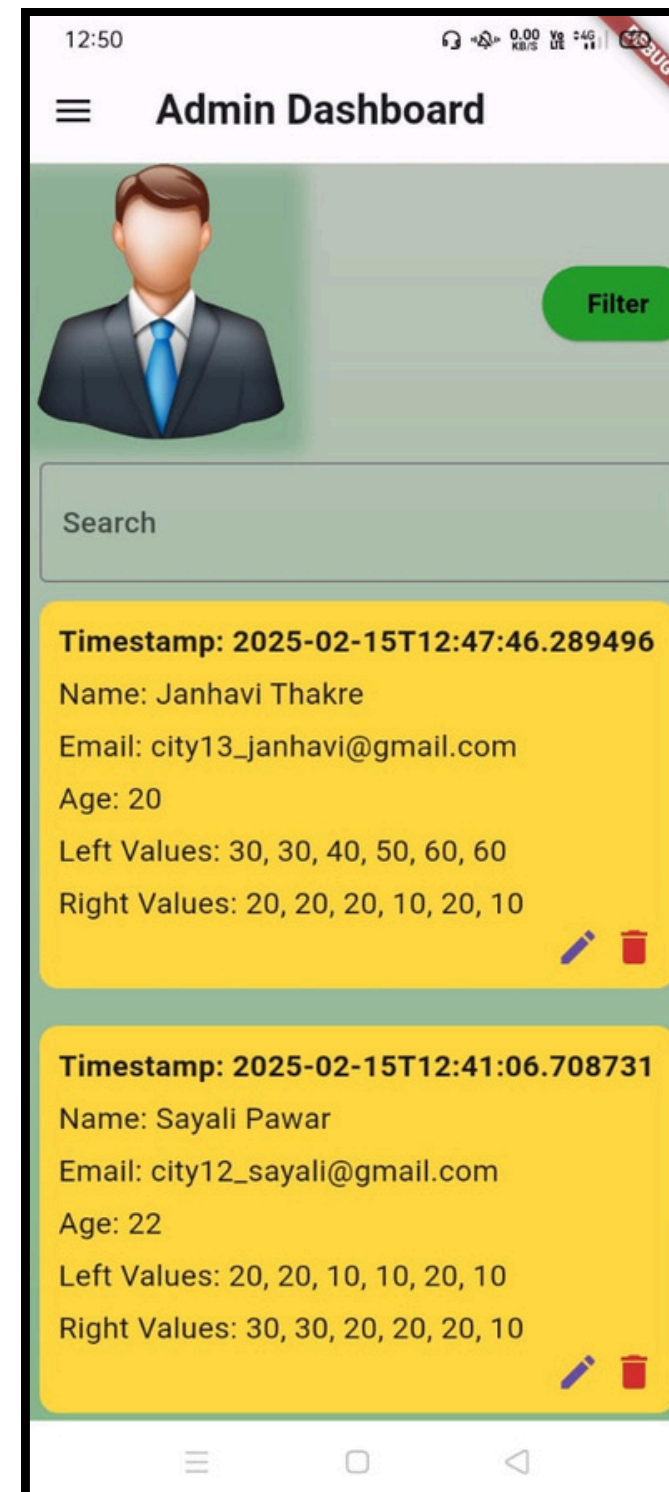
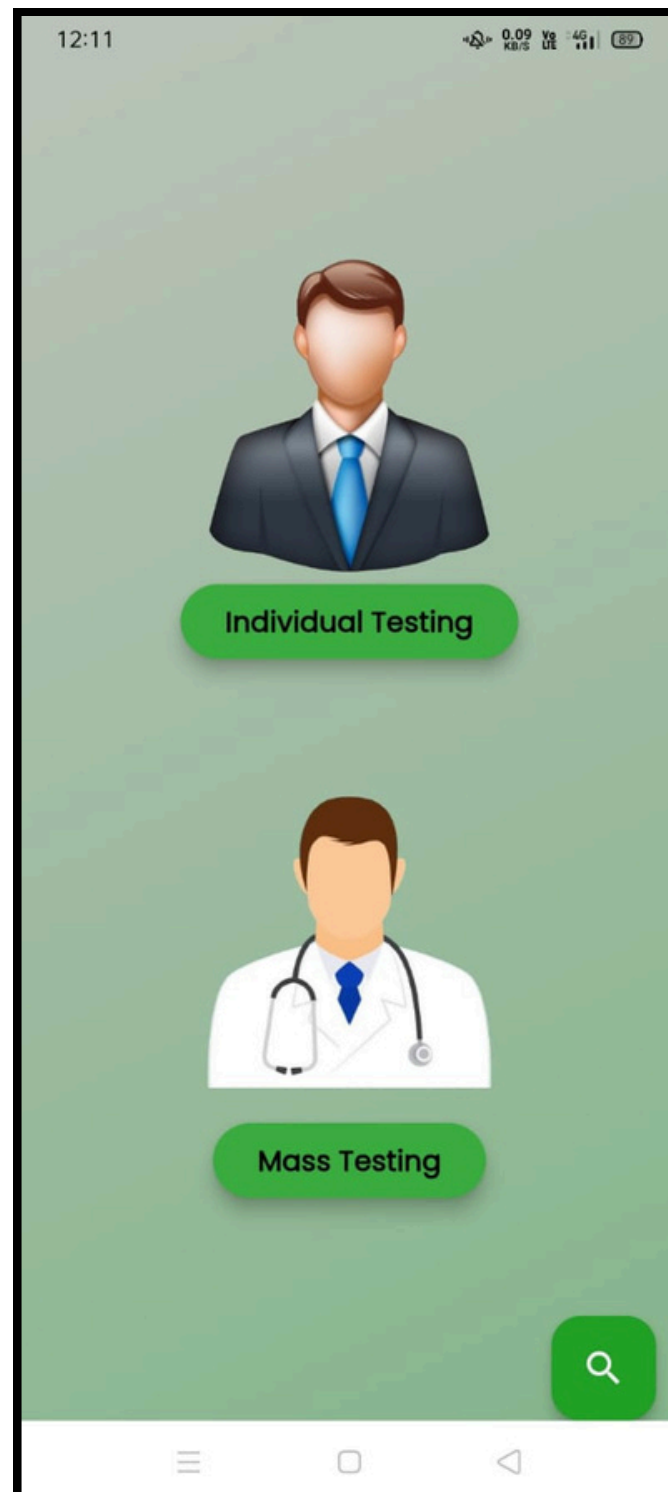
- a. Unit Testing: Verified individual modules like Tone Generator, Noise Calibration, User Feedback, Audiogram Generation.
- b. Integration Testing: Ensured seamless interaction between UI, Firebase, Signal Processing & Audio Modules.
- c. Acceptance Testing: Validated app performance in real-world scenarios by end-users (patients, audiologists).

Results and Discussion

User Interface

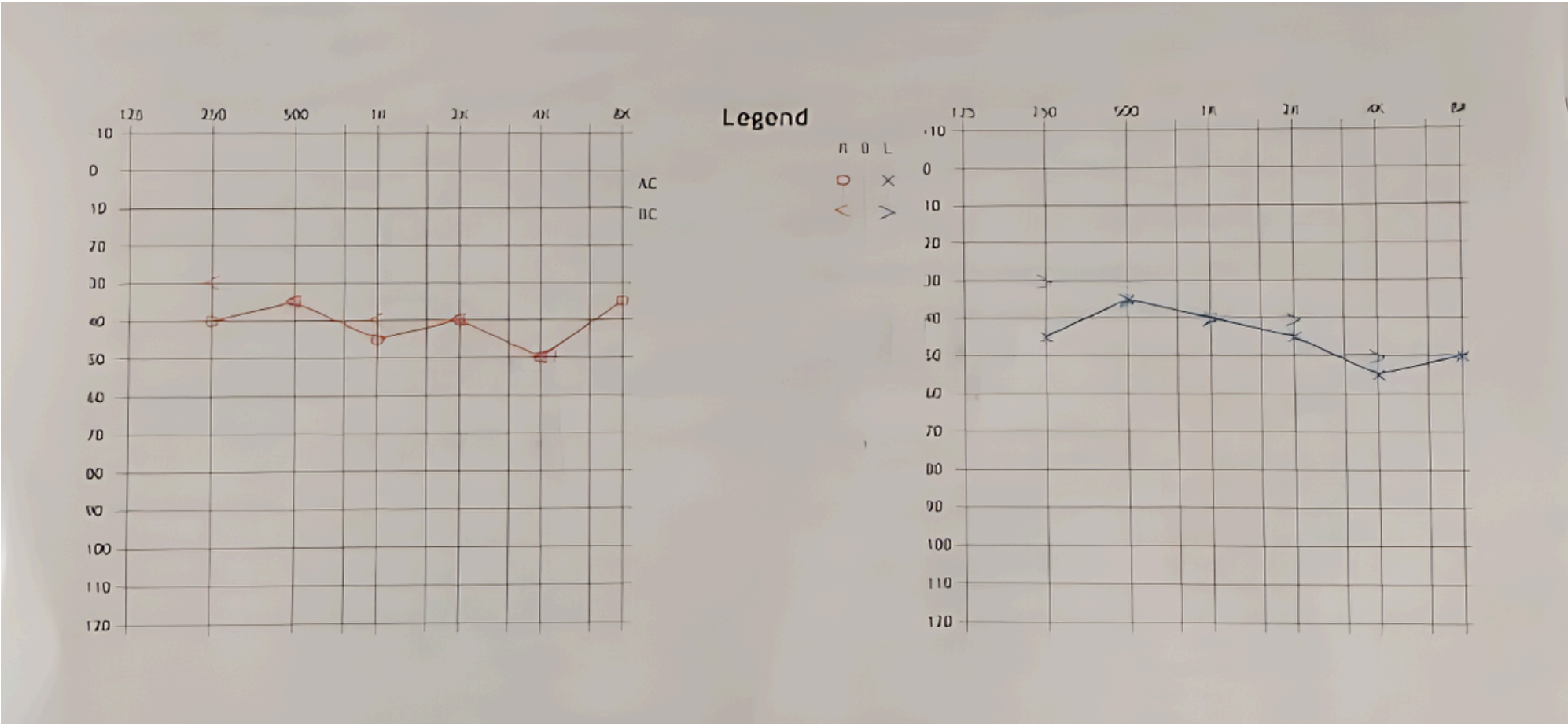


Individual Testing

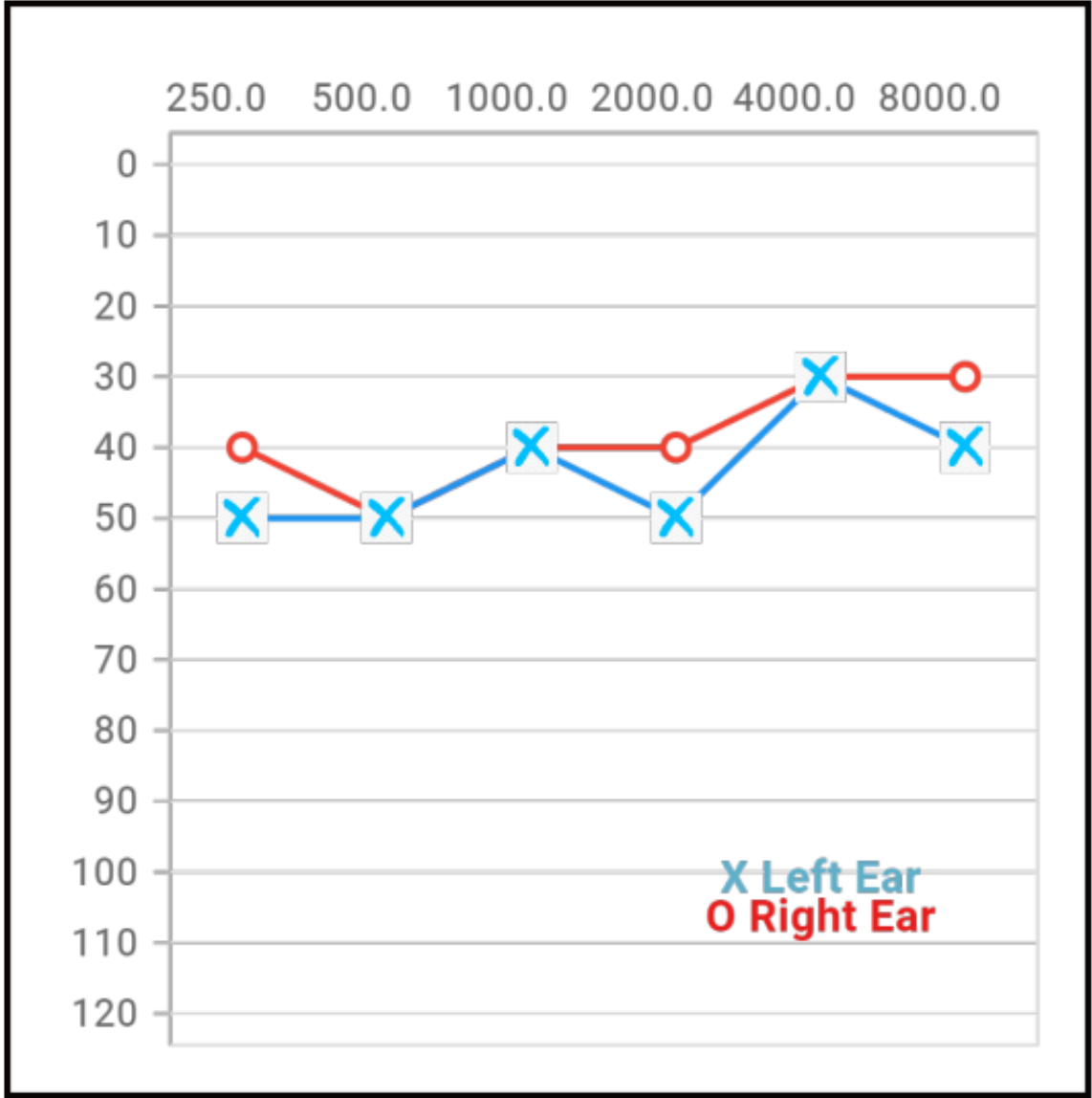


Mass Testing

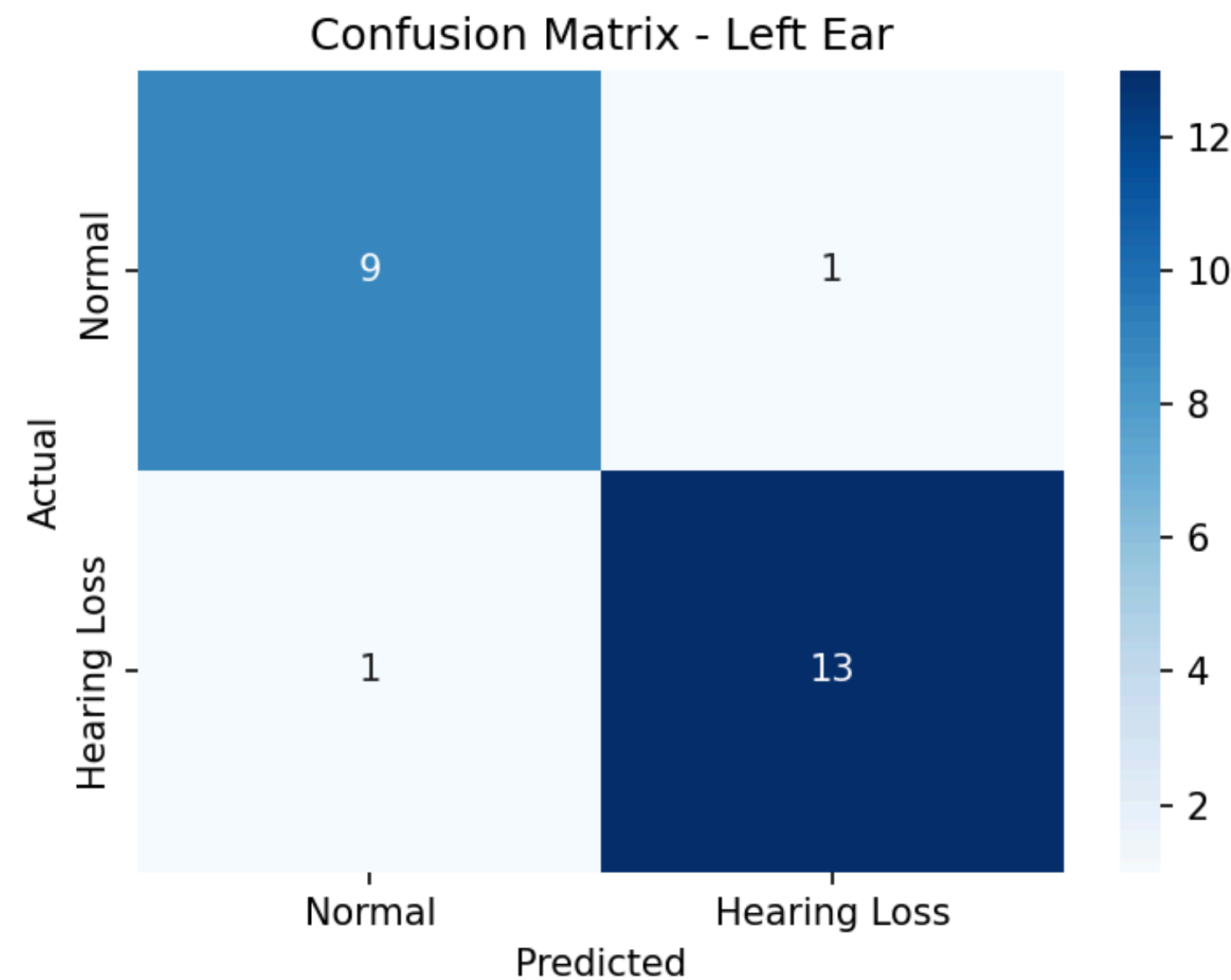
Output of Traditional Method [PTA]



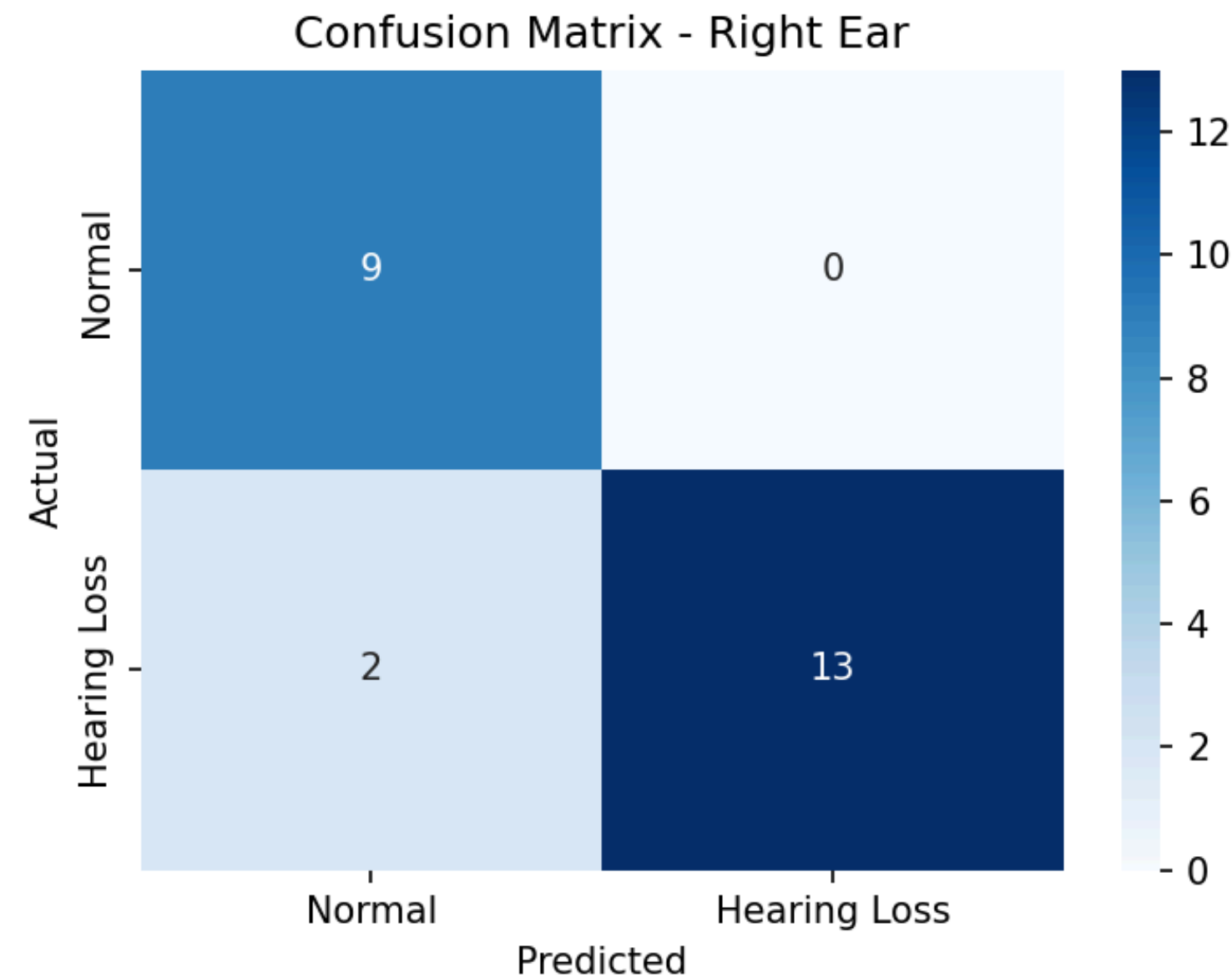
Output of App-Based Audiometer [ABA]



Confusion Matrix

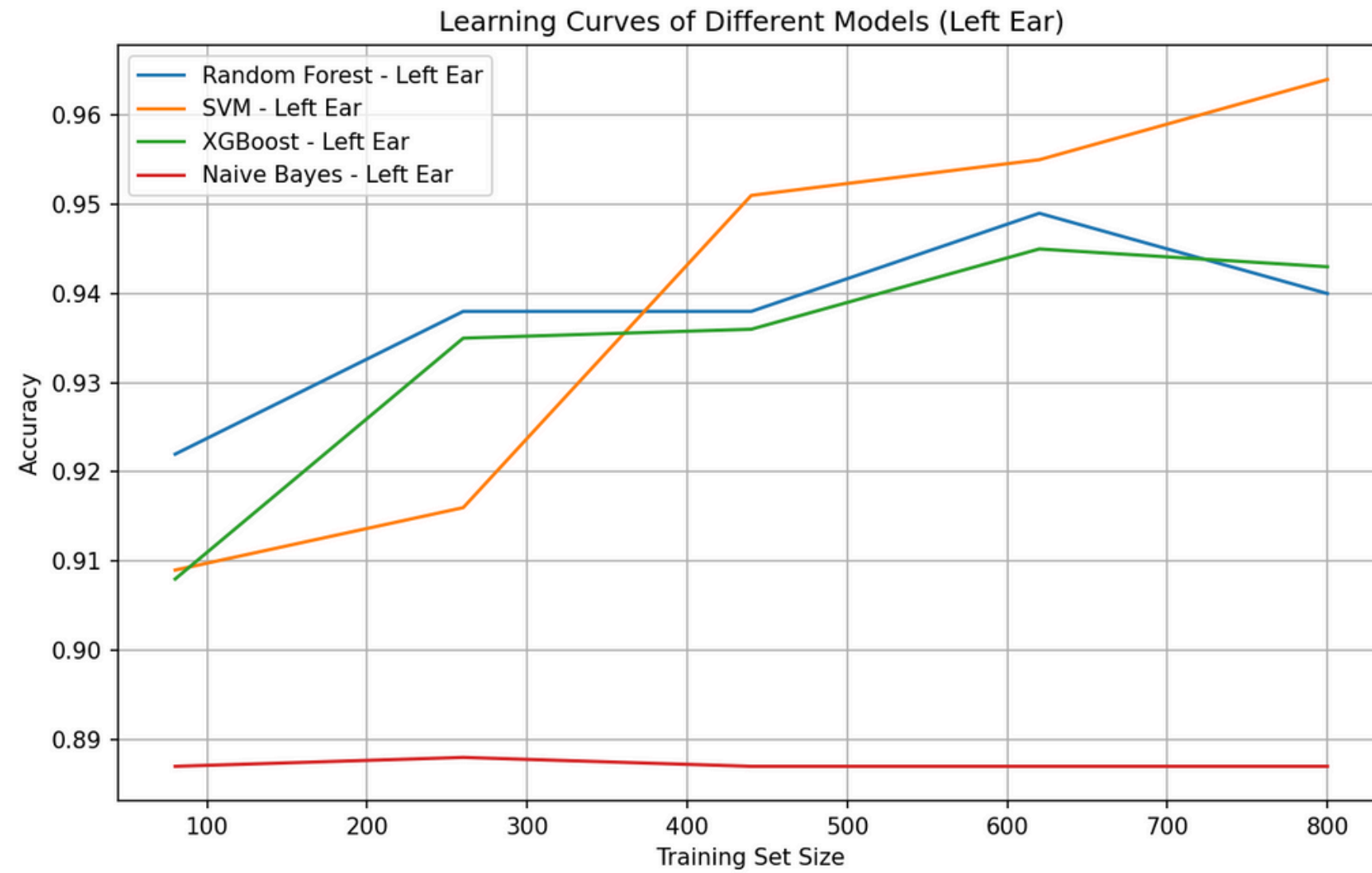


Confusion matrix of Left Ear

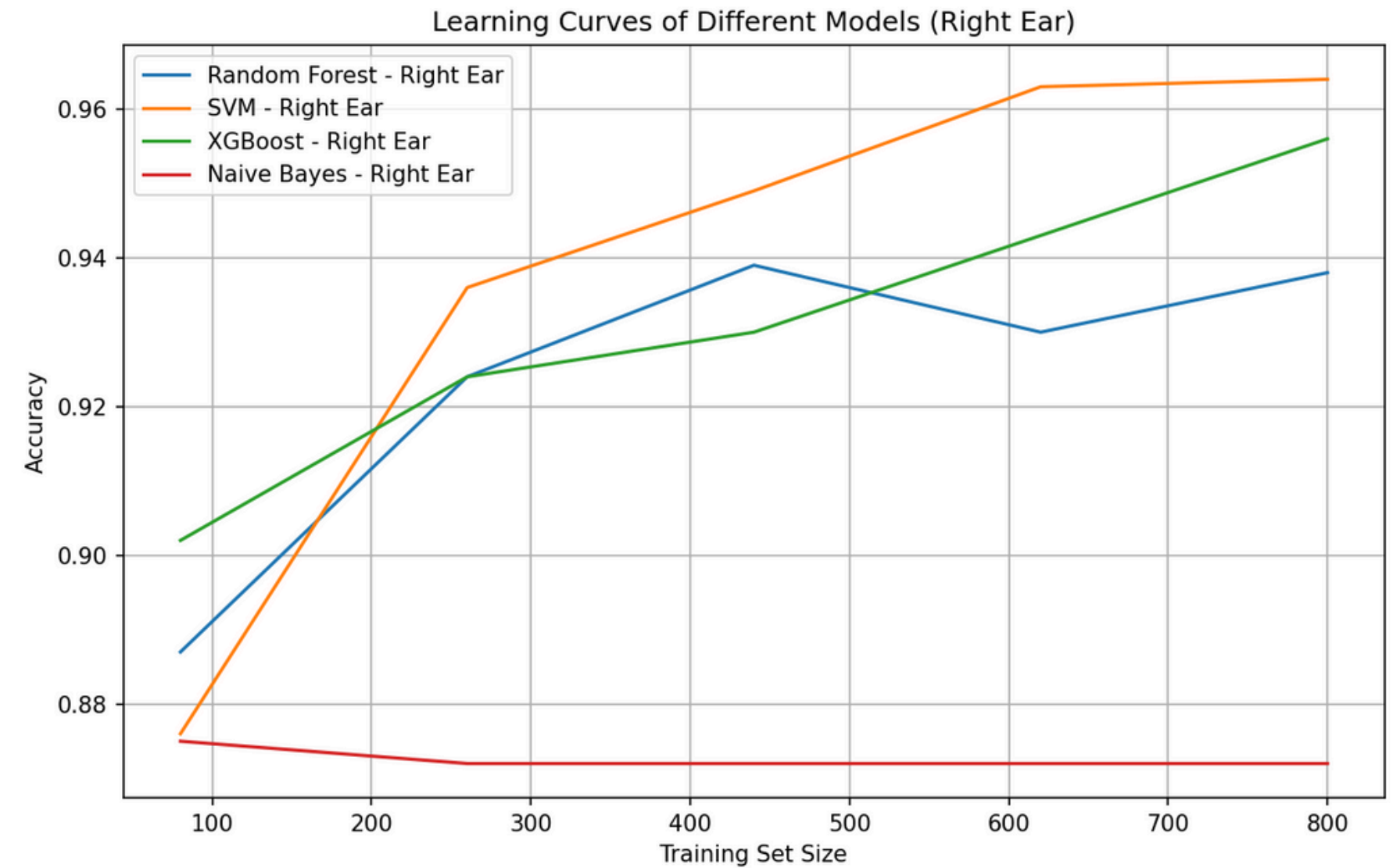


Confusion matrix of Right Ear

Model Result



Left Ear



Right Ear

Model Performance on Left and Right Ear

Model	Left Ear Train Accuracy	Left Ear Test Accuracy	Right Ear Train Accuracy	Right Ear Test Accuracy
Random Forest	0.95	0.94	0.94	0.93
SVM	0.96	0.96	0.97	0.96
XGBoost	0.94	0.93	0.95	0.94
Naive Bayes	0.89	0.89	0.87	0.87

Contribution to Sustainable Development Goals

SDG 3 – Good Health and Well-being

- Facilitates early detection and management of hearing loss, reducing long-term health complications.
- Promotes proactive and inclusive auditory healthcare services.
- Enhances access to affordable and preventive healthcare in underserved and remote areas.
- Supports mental and cognitive development by identifying hearing issues early.
- Helps prevent social isolation and developmental delays linked to undiagnosed hearing impairment.

Conclusion

- This research demonstrates two distinct yet complementary innovations: an App-Based Audiometer and an ML model for hearing loss prediction.
- The App-Based Audiometer provides a cost-effective, portable, and scalable alternative to traditional audiometric testing.
- Data collected from 1000 users via the app was utilized to develop an ML model.
- The ML model, utilizing RFC, Naive Bayes, SVM, and XGBoost, effectively analyzes hearing thresholds and predicts hearing loss severity based on user demographics and audiometric results. The high accuracy of the model highlights its potential for early detection, enabling timely interventions and personalized hearing care.

Paper Publication Details

- Title of Paper: App-Based Audiometer for Hearing Loss Detection and Prediction Using Machine Learning
- Name of International Journal / conference: 9th International Conference on Control Communication, Computing and Automation
- Status : Submitted
- Name of Authors: Vinayak Malavade, Abhishek Bhosale, Aditya Lad, Mohini Mehta, Gaurav Mahale
- Date of paper submission: April 16, 2025
- Indexing: Scopus/WoS/ UGC care

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Thank you